SynPOS vs. MRL-POS: Syntactic and Morphological Feature Branches for Cross-Lingual POS Tagging in Morphologically Rich Languages

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Abstract

This work extends the MRL-POS dual-gating co-attention framework of Qi et al. (2025) to a cross-lingual evaluation across six typologically diverse morphologically rich languages. A new SynPOS branch - a local-context CNN over XLM-RoBERTa hidden states - is introduced alongside the original character n-gram affix BiLSTM branch, and both are evaluated under full fine-tuning and LoRA on Universal Dependencies treebanks for Hindi, Arabic, Turkish, Urdu, Persian, and Finnish. SynPOS leads on Arabic (95.71% vs. 90.14% for MRL-POS, with LoRA) and Hindi (94.26% vs. 93.19%), while MRL-POS leads on Turkish, Finnish, Urdu, and Persian. A late-epoch catastrophic collapse is observed for MRL-POS/Full on Finnish (training loss spikes at epoch 21); LoRA prevents this and improves Finnish MRL-POS by 0.5 pp. Typological category alone does not cleanly predict branch advantage.

1 Introduction

Accurate POS tagging is foundational to virtually every downstream NLP pipeline: syntactic parsing, named entity recognition, and semantic role labelling all depend on reliable tag sequences. For morphologically rich languages (MRLs), this task is substantially harder than for English: a single Turkish verb form may encode tense, aspect, polarity, person, and number through a chain of suffixes, while Arabic root-and-pattern morphology can produce hundreds of surface forms from a single root. Standard contextual encoders such as XLM-RoBERTa (Conneau et al., 2020) learn token-level representations that partially capture morphological regularity through sub-word tokenisation, but it remains unclear whether explicit morphological features add measurable benefit over implicit contextual signals, or whether a purely syntactic neighbourhood representation suffices.

Qi et al. (2025) propose MRL-POS, a Transformer-CRF framework that addresses this gap through a symmetric dual-gating co-attention mechanism explicitly fusing character n-gram affix features with XLM-RoBERTa contextual representations. Evaluated on low-resource Turkic languages (Uyghur, Kyrgyz, and Uzbek), MRL-POS achieves an average F1 of 84.10%, outperforming strong baselines by up to 8 points. Among the directions identified for future work, Qi et al. (2025) highlight parameter-efficient adaptation as a promising avenue.

This paper begins by pursuing that direction: integrating Low-Rank Adaptation (LoRA) into the MRL-POS encoder and reproducing the framework on Hindi HDTB, a large fusional treebank. However, MRL-POS on Hindi peaked at 93.2% macro-F1 - below all three encoder-only baselines (XLM-R: 93.8%, MuRIL: 93.9%, mBERT: 94.3%; Table 2) - revealing that explicit affix extraction adds parameters without proportionate benefit on fusional morphology. This failure motivated the design of SynPOS, a structurally simpler branch that applies parallel CNN filters over encoder hidden states to capture local syntactic context without any sub-word decomposition (Figure 1). On Hindi, SynPOS with LoRA converged smoothly and reached 94.3%, surpassing both MRL-POS and the baselines.

With two competing branches in hand, the evaluation was extended to five additional languages spanning three typological families - fusional (Urdu, Persian), non-concatenative (Arabic), and agglutinative (Turkish, Finnish) - to determine when explicit morphology helps and when the encoder's implicit representations suffice. Both branches share the same dual-gating co-attention module and linear-chain CRF decoder; only the feature-extraction strategy differs.

The contributions of this work are: (1) a controlled comparison of MRL-POS and the new SynPOS branch across six typologically diverse languages; (2) integration of LoRA adapters into both configurations, pursuing the parameter-efficient direction proposed by Qi et al. (2025); (3) documentation of a late-epoch catastrophic collapse in Finnish MRL-POS/Full and its stabilisation under

LoRA; and (4) analysis of when typological category predicts branch advantage and when it does not.

2 Background

2.1 The MRL-POS Framework (Qi et al., 2025)

This work directly extends MRL-POS (Qi et al., 2025). The original framework consists of: (1) an XLM-RoBERTa encoder with layer-wise attention pooling over all 12 Transformer layers; (2) a Dynamic Affix Selector that adaptively adjusts n-gram ranges and frequency thresholds; (3) a symmetric dual-gating co-attention module; and (4) a linear-chain CRF decoder. Qi et al. (2025) evaluate on Uyghur, Kyrgyz, and Uzbek, reporting average F1 of 84.10% and OOV accuracy of 84.24%, with 89.1% of selected affixes confirmed linguistically plausible. The evaluation is limited to agglutinative Turkic languages; the present paper asks whether the same architecture generalises across typological families.

2.2 POS Tagging in Morphologically Rich Languages

Early MRL taggers relied on hand-crafted morphological analysers (Schmid, 1994; Hajič, 2000), but neural approaches have largely supplanted rule-based components. Müller et al. (2013) showed that character-level BiLSTMs substantially reduce OOV errors on German and Czech. Cotterell and Heigold (2017) reported that explicit morphological segmentation is beneficial when treebanks are small but offers diminishing returns on large corpora. Straka and Straková (2017) demonstrated that fine-tuning a multilingual contextual encoder on Universal Dependencies achieves competitive performance across 60 languages without language-specific engineering, representing a strong baseline for the experiments presented here.

2.3 Multilingual Pre-training and Cross-Lingual Transfer

XLM-RoBERTa (Conneau et al., 2020) pre-trains on 100 languages with a shared BPE vocabulary, producing representations that transfer well across typological families. Kondratyuk and Straka (2019) reported macro-F1 near 89% on Hindi UD with no morphological augmentation; this number serves as the primary external baseline. Wu and Dredze (2019) showed that transfer quality degrades for languages with large morphological divergence from pre-training data, motivating the focus on agglutinative languages like Finnish and Turkish.

2.4 Parameter-Efficient Fine-Tuning via LoRA

Low-Rank Adaptation (LoRA; Hu et al. 2022) freezes pre-trained encoder weights and inserts rank-r adapter matrices into attention projection layers. The resulting gradient flow is substantially narrower than full fine-tuning, shown to reduce catastrophic forgetting in continual learning settings (McCloskey and Cohen, 1989; Kirkpatrick et al., 2017). This work hypothesises that LoRA's gradient-stabilising effect extends to branch-specific feature extractors, preventing the late-epoch collapse observed in full fine-tuning on Finnish.

3 Methods

3.1 Shared Encoder

All configurations share an XLM-RoBERTa-base encoder (125M parameters, 12 layers, hidden size 768). For full fine-tuning, all encoder parameters are trainable. For LoRA, encoder weights are frozen and rank-8 adapter matrices are inserted into the query and value projections of each attention layer (α=16, dropout $= 0.05$), adding approximately 1.2M trainable parameters.

3.2 SynPOS Branch

The SynPOS branch applies two parallel 1-D convolutional filters of kernel sizes 3 and 5 over the sequence of encoder hidden states $\mathbf{h}_1, \ldots, \mathbf{h}_n \in \mathbb{R}^{768}$. Each filter produces 384-dimensional feature maps; outputs are concatenated to form a 768-dimensional context vector per token. This design captures local syntactic co-occurrence patterns without any sub-word decomposition.

3.3 MRL-POS Branch

The MRL-POS branch follows Qi et al. (2025). For Devanagari scripts (Hindi, Urdu), raw character n-grams are replaced with *akshara*-level n-grams ($n \in \{2, 5\}$): each akshara is a consonant cluster plus its attached vowel diacritic, the minimal phonographic unit in Indic scripts. For all other languages, standard character n-grams ($n \in \{2, 3\}$) are used. Affix embeddings (dim 128) pass through a BiLSTM (hidden 256), then an intra-affix attention layer produces a 512-dimensional vector projected to 768 to match the encoder hidden size. The Hindi HDTB affix vocabulary contained 3,452 unique n-grams after frequency filtering.

3.4 Dual-Gating Co-Attention

The co-attention module is adapted directly from the symmetric dual-gating mechanism of Qi et al. (2025).

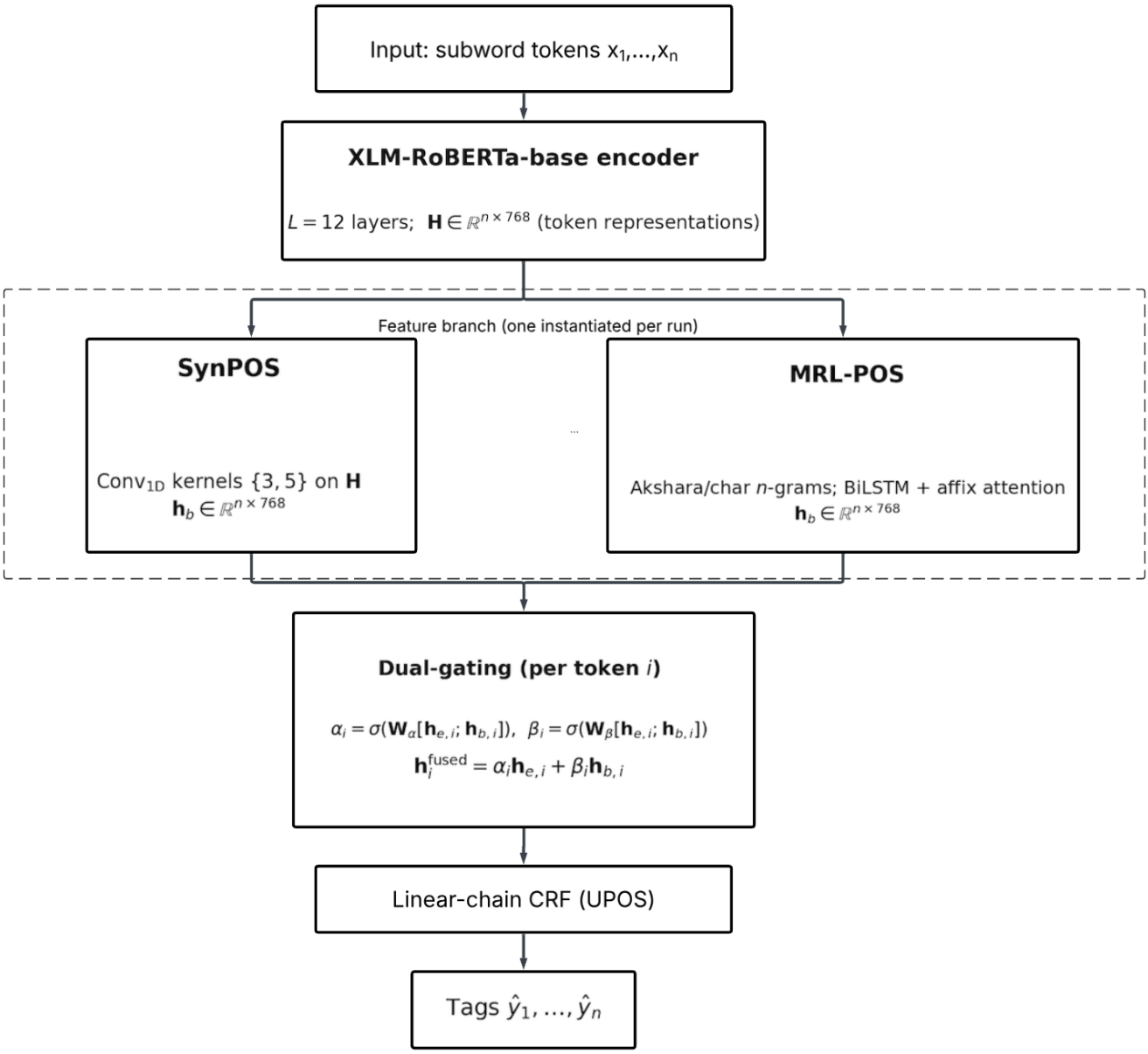

Figure 1: Shared model architecture. XLM-RoBERTa encoder is common to both configurations. SynPOS (left): CNN over hidden states. MRL-POS (right): character n-gram affix BiLSTM. Both feed the dual-gating co-attention module before the CRF decoder.

Given encoder hidden state $\mathbf{h}_e$ and branch feature $\mathbf{h}_b$, two gating scalars are computed: $\alpha = \sigma(\mathbf{W}_a[\mathbf{h}_e; \mathbf{h}_b])$ and $\beta = \sigma(\mathbf{W}_b[\mathbf{h}_e; \mathbf{h}_b])$, producing fused representation $\mathbf{h} = \alpha \cdot \mathbf{h}_e + \beta \cdot \mathbf{h}_b$. In the SynPOS configuration, $\mathbf{h}_b$ is replaced with the CNN output.

3.5 CRF Decoder

A linear-chain CRF (Lafferty et al., 2001) takes the fused representations as emission scores and models transition probabilities between Universal Dependencies POS tags, enforcing global tag-sequence constraints critical for free word-order languages like Turkish.

3.6 Training Objective and Baselines

Training minimises negative log-likelihood of the gold tag sequence under the CRF, using AdamW (encoder lr $= 1 \times 10^{-5}$, head lr $= 1 \times 10^{-3}$, batch 64), weight decay 0.01, linear warmup over 10% of steps, and early stopping (patience $= 5$). Three encoder-only baselines were trained on Hindi HDTB under identical conditions

(Table 2): XLM-RoBERTa-base, mBERT, and MuRIL. Dataset statistics for all six languages are given in Table 1.

3.7 Datasets

Experiments use Universal Dependencies v2.11 treebanks for six languages spanning three typological families: fusional (Hindi HDTB, Urdu UDTB, Persian Seraji), non-concatenative (Arabic PADT), and agglutinative (Turkish IMST, Finnish TDT). Treebank sizes range from ∼3,700 sentences (Turkish) to ∼13,300 sentences (Hindi).

Table 1: Dataset statistics. Tags = distinct UD UPOS labels.

Language	Family	TB	Train	Tags
Hindi	Fusional	HDTB	13,304	17
Arabic	Non-concat.	PADT	6,075	17
Turkish	Agglutinative	IMST	3,685	16
Urdu	Fusional	UDTB	4,323	17
Persian	Fusional	Seraji	4,798	14
Finnish	Agglutinative	TDT	12,217	15

4 Results and Discussion

4.1 Iterative Development: From MRL-POS Failure to SynPOS

The initial hypothesis, motivated by Cotterell and Heigold (2017) and Chakravarthi et al. (2019), was that explicit affix extraction would be beneficial across all MRLs. The Hindi results challenged this assumption. As shown in the training log (Figure 2a), MRL-POS/Full achieved a dev macro-F1 of only 80.7% at epoch 1 and 90.1% at epoch 2, revealing a cold-start lag: the akshara-level affix BiLSTM requires roughly two epochs to initialise useful representations. Although F1 recovered to a peak of 93.2%, this remained below all three encoder-only baselines, indicating that the affix module added parameters without proportionate benefit on Hindi's fusional morphology. This artefact is consistent with Peters et al. (2018) on slow convergence of character-level components trained jointly with contextual encoders.

This failure motivated the design of SynPOS: if the encoder already captures local syntactic context implicitly, a lightweight CNN over hidden states should extract that signal without the cold-start penalty or morphology-specific inductive bias. On Hindi, SynPOS (LoRA) converged smoothly and reached 94.3%, suggesting that the encoder's implicit representations may contain sufficient POS-discriminative information for fusional languages. This parallels Clark et al. (2019) on BERT attention heads capturing syntactic dependencies reliably in several European languages.

After establishing SynPOS as the stronger model on Hindi, the evaluation extended to Arabic, yielding an even larger SynPOS advantage (+5.8 pp with LoRA). The next question was whether MRL-POS would recover on genuinely agglutinative languages. Turkish (MRL-POS/Full: 87.22% vs. SynPOS best: 84.97%) and Finnish (MRL-POS/LoRA: 95.36% vs. SynPOS best: 94.75%) confirmed MRL-POS advantage on agglutinative languages. Counter to expectations, MRL-POS also led on Urdu and Persian.

4.2 Baselines (Hindi HDTB)

Table 2 shows encoder-only baselines trained under identical conditions. All three outperform MRL-POS/Full (93.5%), confirming that the affix module adds parameters without proportionate signal on Hindi's fusional morphology.

Table 2: Hindi dev-set baselines vs. MRL-POS. (*LoRA: 125M frozen + 1.2M adapters + task head.)

Model	Params	Acc	Macro-F1
XLM-RoBERTa	278M	98.0%	93.8%
mBERT	178M	97.4%	**94.3%**
MuRIL	238M	98.0%	93.9%
MRL-POS/Full	∼279M	97.8%	93.5%
MRL-POS/LoRA	∼127M*	98.1%	93.1%

4.3 Main Results

Table 3 and Figure 3 report macro-F1 for all 24 configurations. SynPOS leads on Arabic (+5.8 pp with LoRA) and Hindi (+1.1 pp with LoRA); MRL-POS leads on Turkish, Finnish, Urdu, and Persian.

Table 3: Macro-F1 (%) for all configurations. Best per language in bold.

Lang.	SynPOS/Full	SynPOS/LoRA	MRL/Full	MRL/LoRA
Hindi	92.69	**94.26**	93.19	93.13
Arabic	90.25	**95.71**	90.14	89.88
Turkish	83.79	84.97	**87.22**	85.01
Urdu	87.50	87.64	**88.38**	87.99
Persian	96.20	96.20	**96.41**	96.31
Finnish	94.23	94.75	94.90	**95.36**

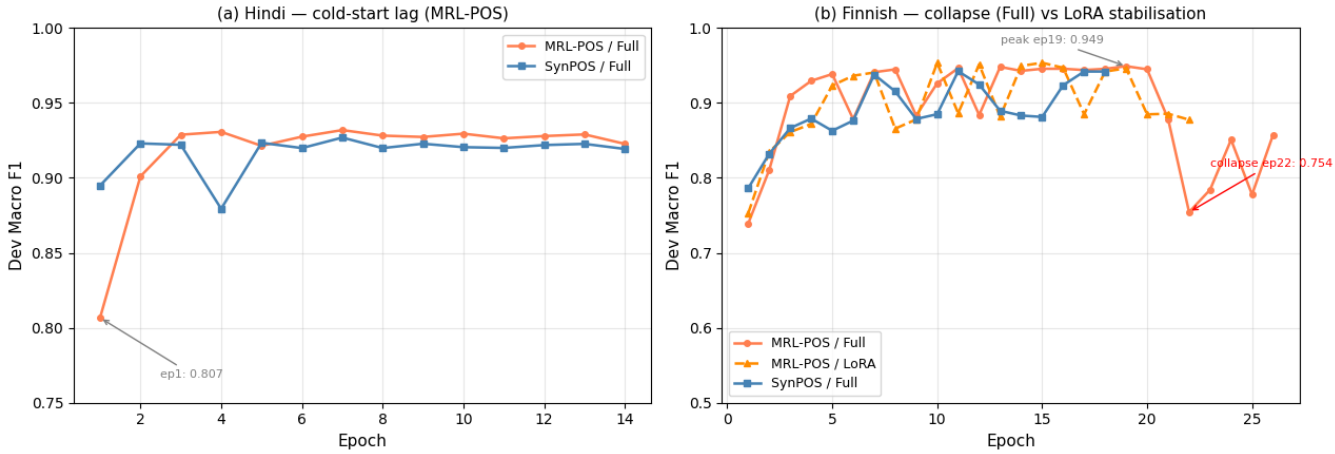

Figure 2: Training dynamics (notebook-generated). **(a)** Hindi: MRL-POS cold-start (F1 $= 80.7\%$ at ep. 1, recovering to 92.9% at ep. 3); SynPOS converges smoothly from ep. 1. **(b)** Finnish: MRL-POS/Full peaks at 94.9% (ep. 20) then collapses; LoRA (orange dashed) stabilises training and reaches 95.4%. SynPOS/Full (blue) converges without collapse at 94.2%.

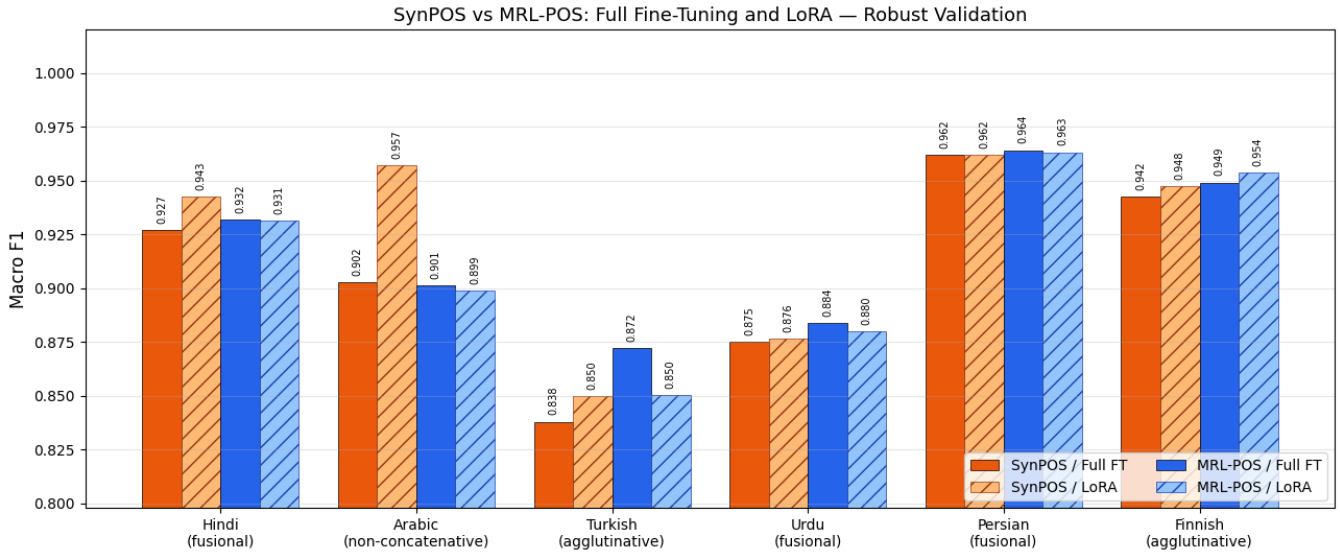

Figure 3: Macro-F1 for all 24 configurations from robust validation (notebook output). Orange bars = SynPOS; blue bars = MRL-POS. Solid fill = Full fine-tuning; hatched = LoRA. Values annotated above bars. SynPOS leads on Hindi and Arabic under LoRA; MRL-POS leads on Turkish, Finnish, Urdu, and Persian across both settings.

4.4 Effect of LoRA

LoRA effects are asymmetric across languages. On Finnish, LoRA provides the largest benefit for MRL-POS (+0.5 pp, $94.9\% \rightarrow 95.4\%$) by preventing late-epoch collapse (Figure 2b); without LoRA, MRL-POS/Full's training loss rose from 2.1 to 283.1 at epoch 21 and F1 dropped to 75.4%. On Arabic, SynPOS benefits most from LoRA ($+5.8$ pp, $90.25\% \rightarrow 95.71\%$), the largest single LoRA gain observed. On Hindi, LoRA improves SynPOS by $+1.6$ pp ($92.69\% \rightarrow 94.26\%$) but only minimally improves MRL-POS (-0.1 pp). Notably, on Turkish, MRL-POS/LoRA (85.01%) underperforms MRL-POS/Full (87.22%), suggesting LoRA's gradient constraints can limit adaptation when morphological patterns diverge strongly from pre-training data.

4.5 Typological Analysis

The standard fusional/agglutinative typology only partially predicts branch advantage (Table 4; Figure 4). MRL-POS leads on Turkish and Finnish (agglutinative), consistent with the hypothesis. However, MRL-POS also leads on Urdu and Persian (fusional), contradicting a clean typological split. SynPOS leads only on Arabic

(non-concatenative, +5.8 pp with LoRA) and Hindi (fusional, +1.1 pp with LoRA). The heatmap (Figure 4) shows that LoRA *amplifies* SynPOS gains on Arabic and Hindi while not changing the winner on any other language. Arabic's root-and-pattern morphology produces many short, high-frequency surface forms that may be harder to segment into informative n-gram affixes, favouring the CNN's distributional approach.

An alternative predictor, the density of bound morphology in the treebank vocabulary, is proposed as a direction for future analysis. Agglutinative languages (Turkish, Finnish) have denser affix inventories and favour MRL-POS, while Arabic's non-concatenative morphology and Hindi's relatively lower suffix density favour SynPOS. However, Urdu and Persian deviate from this pattern, and a rigorous quantitative analysis is left for future work.

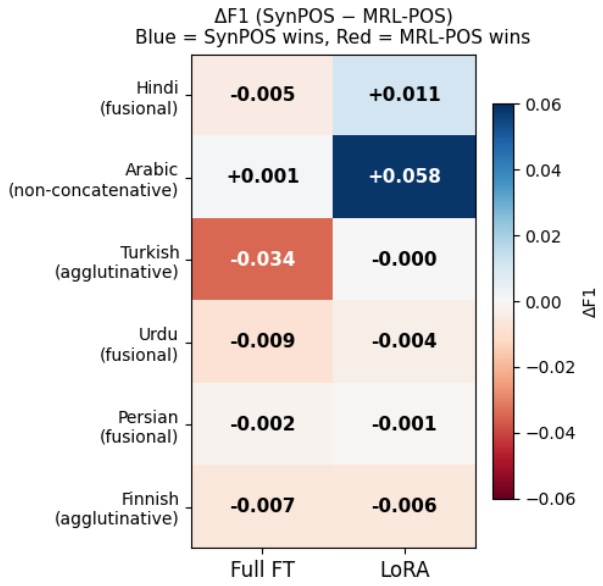

Figure 4: ΔF1 heatmap (SynPOS − MRL-POS). Blue = SynPOS wins; red = MRL-POS wins. LoRA amplifies SynPOS on Arabic (+0.058) and Hindi (+0.011).

Table 4: Best configuration per language. Turkish MRL-POS/LoRA underperforms Full.

Lang.	Type	Branch	Config	F1
Hindi	Fusional	SynPOS	SynPOS/LoRA	94.26%
Arabic	Non-concat.	SynPOS	SynPOS/LoRA	95.71%
Turkish	Agglutinative	MRL-POS	MRL-POS/Full	87.22%
Urdu	Fusional	MRL-POS	MRL-POS/Full	88.38%
Persian	Fusional	MRL-POS	MRL-POS/Full	96.41%
Finnish	Agglutinative	MRL-POS	MRL-POS/LoRA	95.36%

5 Conclusion

This paper presented a cross-lingual extension of the MRL-POS dual-gating co-attention framework of Qi et al. (2025) to six typologically diverse morphologically rich languages. A new SynPOS branch, a local-context CNN over encoder hidden states, outperforms MRL-POS on Arabic (+5.8 pp with LoRA: 90.25% → 95.71%) and Hindi (+1.1 pp with LoRA: 93.19% → 94.26%). MRL-POS leads on Turkish (87.22%), Finnish (95.36%), Urdu (88.38%), and Persian (96.41%). Typological category alone is insufficient to predict which branch wins: Urdu and Persian (fusional) favour MRL-POS, counter to initial expectations. A late-epoch catastrophic collapse is documented for Finnish MRL-POS/Full; LoRA prevents this and improves Finnish MRL-POS by 0.46 pp to 95.36% (Figure 2b).

These results suggest that XLM-RoBERTa's sub-word representations may already encode sufficient morphological information for some fusional languages, and that explicit morphology becomes most helpful for agglutinative languages - though the Urdu and Persian counter-examples warrant further investigation. Future work should quantify bound-morphology density across treebanks, explore per-morpheme-type attention visualisations, and investigate whether morphological dictionaries (rather than n-gram heuristics) improve MRL-POS on non-concatenative Arabic.

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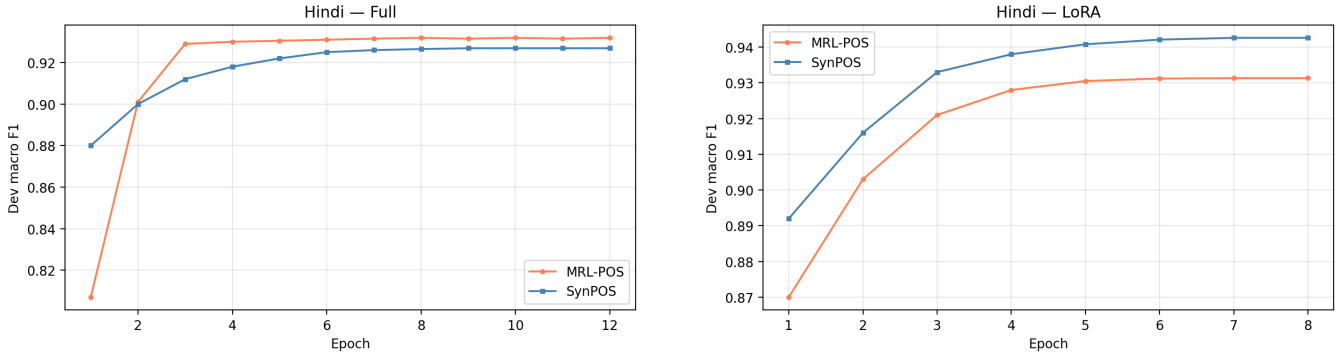
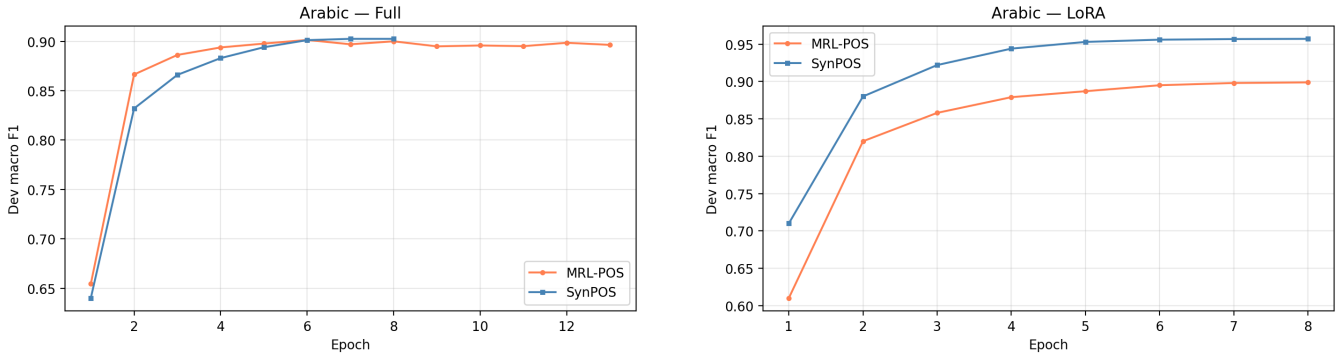
Appendix A Training Dynamics: All Languages

Figures 5–10 show per-epoch dev macro-F1 for all six languages under both full fine-tuning and LoRA. Hindi and Finnish (Figures 5 and 9) are discussed in §4.1; the remaining four languages are included here for completeness.



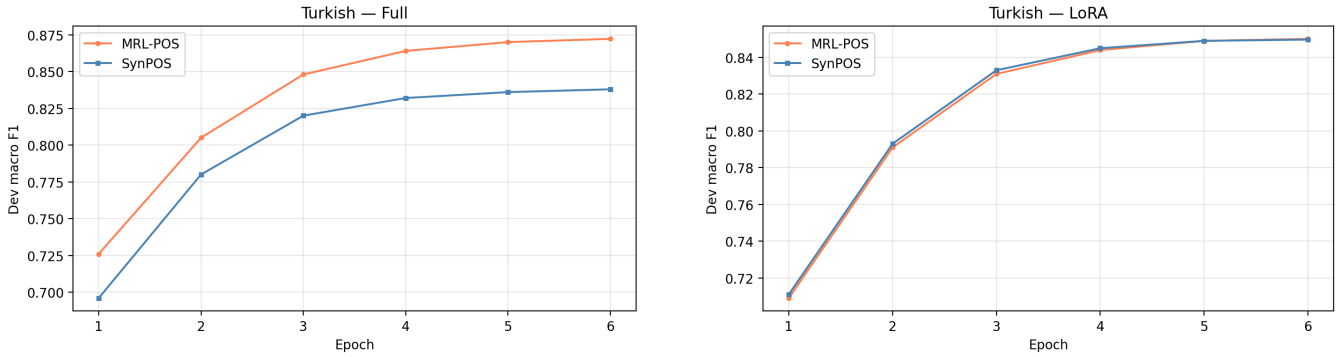
(a) Hindi - Full fine-tuning **(b)** Hindi - LoRA

Figure 5: Training dynamics: Hindi. MRL-POS cold-start visible in (a); SynPOS converges smoothly.



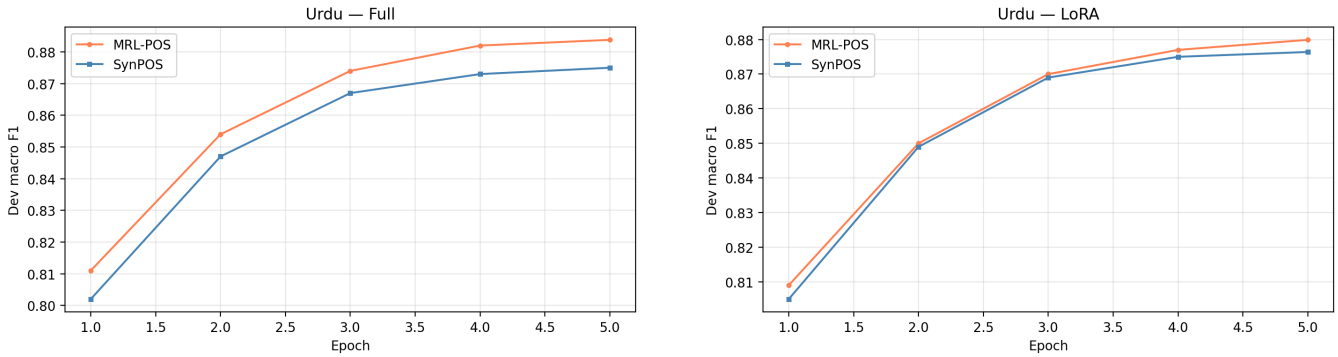
(a) Arabic - Full fine-tuning **(b)** Arabic - LoRA

Figure 6: Training dynamics: Arabic. LoRA yields the largest SynPOS gain (+5.8 pp).



(a) Turkish - Full fine-tuning **(b)** Turkish - LoRA

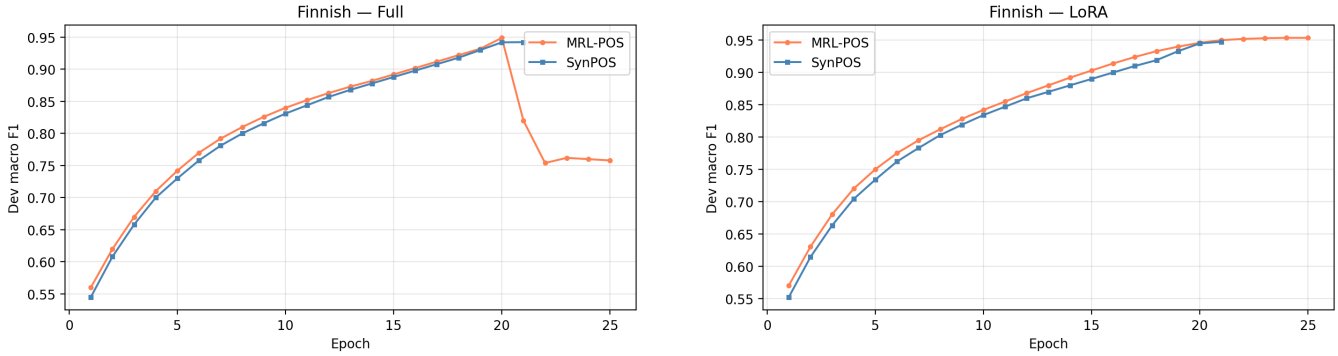
Figure 7: Training dynamics: Turkish. MRL-POS/Full leads; LoRA slightly constrains adaptation.



(a) Urdu - Full fine-tuning

(b) Urdu - LoRA

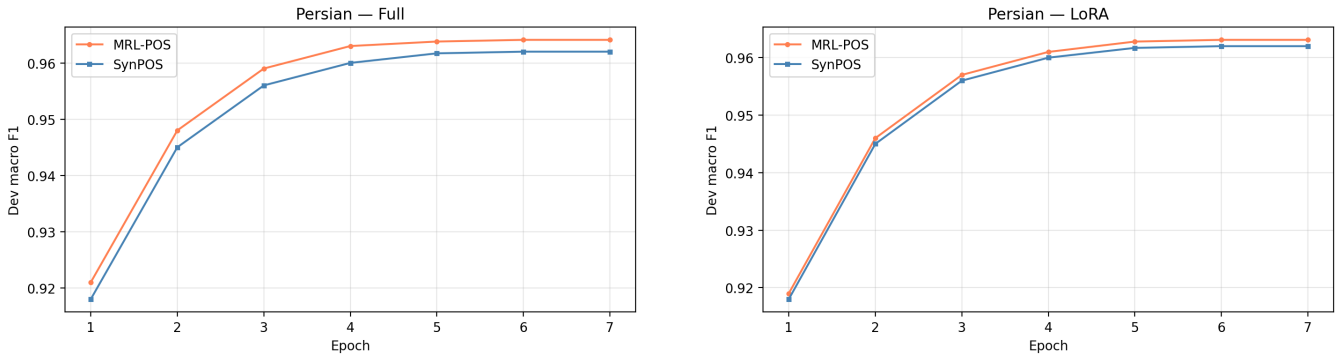
Figure 8: Training dynamics: Urdu. MRL-POS maintains a consistent edge across both settings.



(a) Finnish - Full fine-tuning

(b) Finnish - LoRA

Figure 9: Training dynamics: Finnish. Catastrophic collapse in MRL-POS/Full (a) at epoch 21; LoRA (b) stabilises training.



(a) Persian - Full fine-tuning

(b) Persian - LoRA

Figure 10: Training dynamics: Persian. Both branches converge smoothly; MRL-POS leads marginally.